

PRODUCT REQUIREMENTS DOCUMENT

Contextual Review Routing

Separating OTA-caused friction from genuine hotel experience scores

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1. Executive Summary

Agoda scores 0.2–0.5 points lower than Booking.com for identical hotels across 10 luxury properties in Bali, Singapore, and Tokyo. SAS-validated analysis (N=10 hotels, paired t-test: $t=-11.04$, $p<.0001$) confirms this gap is large, statistically robust, and consistent in direction across every hotel tested.

Sampling bias (H1) — supported by the data: Agoda's reviewer base is 65.1% Asian vs. Booking.com's 16.9% (nationality data is now captured on both platforms). Controlling for hotel and platform across 1,850 reviews, reviewer nationality significantly predicts score ($p=0.027$); once nationality is accounted for, platform itself has virtually no remaining effect ($p=0.89$). The aggregate score gap is best explained by *who reviews on each platform*, not by a platform-caused experience difference.

OTA-caused friction (H3) — plausible, not statistically confirmed: review text patterns (booking confirmation issues, payment holds, check-in problems) suggest this mechanism occurs, but it was not directly tested against the aggregate gap in this analysis. It remains a secondary, qualitative hypothesis, independent of the sampling-bias finding above.

Secondary finding (H2): Booking.com reviews show significantly higher score variance than Agoda's (Folded F test, $p<.0001$) — a separate, confirmed pattern worth further study.

This PRD still proposes Contextual Review Routing. The rationale shifts slightly given the H1 finding: regardless of what explains the aggregate gap, individual hotels experience real per-platform scoring differences that affect partner trust and allocation decisions. Structurally separating any genuinely OTA-caused complaints from hotel scores remains good practice for protecting hotel relationships — but this feature should not be assumed to close most of the measured gap, since composition (H1), not friction (H3), is the data-confirmed primary driver. We recommend pairing this feature with direct testing of H3 (e.g., structured complaint tagging or NLP topic modeling on review text) before treating it as the primary lever.

2. Problem Statement

2.1 Background

Agoda operates a merchant model — it controls the full payment and booking confirmation flow, unlike Booking.com which uses an agency model where the hotel handles most of the experience chain. Every platform failure (a lost confirmation email, a payment hold at check-in, a cancellation the hotel was never notified of) can become a touchpoint the traveler experiences as part of their hotel stay, and may get scored against the hotel rather than the platform.

However, this analysis's primary finding is that the current Agoda-Booking.com score gap is substantially explained by reviewer composition differences (H1), not by this friction mechanism (H3). Both can be true simultaneously — composition may explain most of the aggregate gap while friction still degrades individual hotel relationships in specific cases — but the data does not support claiming H3 as the primary explanation for the measured score deflation.

2.2 Evidence from Data

Hotel	City	Agoda	Booking	Gap	Risk Tier
Four Seasons Resort Bali at Sayan	Bali	9.1	9.5	−0.4	At-Risk

The St. Regis Bali Resort	Bali	9.2	9.7	−0.5	At-Risk
Alila Villas Uluwatu	Bali	9.0	9.4	−0.4	At-Risk
Marina Bay Sands	Singapore	8.8	9.2	−0.4	At-Risk
Raffles Singapore	Singapore	9.2	9.7	−0.5	At-Risk
Mandarin Oriental Singapore	Singapore	9.0	9.3	−0.3	Watch
Park Hyatt Tokyo	Tokyo	8.9	9.3	−0.4	At-Risk
Mandarin Oriental Tokyo	Tokyo	9.1	9.3	−0.2	Watch
The Peninsula Tokyo	Tokyo	9.1	9.3	−0.2	Watch
The Ritz-Carlton Tokyo	Tokyo	8.8	9.2	−0.4	At-Risk

Sub-score analysis (averaged across all 10 hotels) shows the gap concentrates most in **Room Comfort (−0.41)**, followed by Location (−0.37) and Staff Performance (−0.33). Cleanliness shows the smallest gap (−0.14). As detailed above, this pattern is now understood to correlate with reviewer nationality composition rather than being independently diagnostic of hotel quality issues.

3. User Insight

The post-booking journey on Agoda currently looks like this:

- Book → Confirmation email (ambiguous language, sometimes delayed)
- No proactive check-in instructions sent to traveler
- Arrival → Hotel sometimes has no record of the booking
- Dispute → Agoda CS unreachable or slow to resolve
- Review prompt (24–48 hrs post-checkout) → Frustrated traveler scores hotel for Agoda's failure

This journey remains a real, qualitatively observed pattern in review text, independent of the quantitative H1 finding above. The review prompt is the only structured feedback mechanism; without a way to separate any OTA-caused issues from hotel quality, that frustration flows into the hotel's score.

4. Proposed Solution

4.1 Feature: Contextual Review Routing

Before a user submits their hotel rating, Agoda surfaces a single pre-question:

“Before you rate — was your experience affected by any of the following?”

☐ Booking confirmation issue
 ☐ Payment or refund problem
 ☐ Check-in process (not the hotel's fault)
 ☐ None of the above — my experience was about the hotel itself

4.2 Routing Logic

- If no box is checked → rating flows to the hotel score as normal.
- If any OTA-cause box is checked → the complaint routes to a new Agoda Service Score (visible on the platform; not attributed to the hotel). A CS ticket is auto-created for the flagged issue.
- Hotel score only captures genuine hospitality experience.

5. Success Metrics

Metric	Target	Timeframe
Agoda-Booking.com score gap	Reduce residual (non-composition) gap by 0.05–0.10 pts	6 months post-launch
Hotel re-allocation rate	At-Risk → Safe/Watch migration for 40% of at-risk hotels	9 months post-launch
Agoda Service Score complaints resolved	> 80% resolved within 48 hrs	Ongoing
H3 validation study	Complete structured complaint-tagging analysis	3 months post-launch

Note the score-gap target above is intentionally modest relative to the full 0.37-point measured gap, since the majority of that gap is attributed to reviewer composition (H1), which this feature does not address. Success should be measured against the smaller, friction-attributable share of the gap.

6. Trade-offs & Risks

UPSIDE

Even a partial fix to genuinely OTA-caused complaints improves specific hotel relationships and removes a source of partner frustration, independent of its size relative to the total gap.

DOWNSIDE

Agoda now owns a visible Service Score exposing its own operational failures, and the expected impact on the headline score gap is smaller than originally assumed, since most of the gap is compositional rather than friction-driven. This should be communicated clearly to stakeholders to avoid overpromising impact.

RISK: GAMING

Hotels may coach guests to check OTA-cause boxes to deflect legitimate complaints. Mitigation: only surface routing options when Agoda's own system logs show a confirmed issue event (CS contact, payment hold flag, confirmation resend) — not based on guest self-report alone.

7. Scope

In Scope

- Pre-rating modal with OTA-cause checkboxes
- Routing logic: checked = Agoda Service Score + CS ticket; unchecked = hotel score
- Agoda Service Score display on hotel listing page
- A follow-up quantitative study (H3) using structured complaint tagging or NLP on review text, to test the friction hypothesis directly before scaling further investment

Out of Scope

- Retroactive score correction for existing reviews
- Changes to the overall rating display scale
- Any claim that this feature will close the majority of the measured score gap

8. Dependencies

- CS team must be staffed to handle auto-created tickets from routing
- Data team must build Agoda Service Score infrastructure and the H3 validation study pipeline
- Legal must review whether the routing mechanism creates any implicit liability admission
- Supply Partnerships team must brief hotel partners honestly: this feature addresses a real but partial share of the score gap, with the larger share attributed to reviewer composition, not Agoda's platform

9. Analytical Foundation

This PRD is grounded in analysis of 10 luxury hotels (Bali, Singapore, Tokyo) using Python (Apify-scraped review data) and SAS Studio (statistical validation, PROC GLM with hotel fixed effects). Key outputs:

- Paired t-test (N=10 hotels): mean gap = -0.37 , $t=-11.04$, $p<.0001$
- Reviewer profile: 65.1% Asian on Agoda vs. 16.9% on Booking.com; nationality data ~100% captured on both platforms in this scrape
- Cross-platform regression (N=1,850, hotel fixed effects): nationality significantly predicts score ($p=0.027$); platform itself does not ($p=0.89$) once nationality is controlled — H1 supported
- Score variance (H2): Booking.com shows significantly higher variance than Agoda (Folded F test, $p<.0001$)
- Sub-score analysis: Room Comfort (-0.41), Location (-0.37), Staff Performance (-0.33) are the largest gaps; Cleanliness smallest (-0.14)
- BA revenue model: 7 At-Risk hotels $\times$ \$3,456/year = \$24,192 annual GMV at risk

Full SAS output, Python cleaning scripts, and raw CSV files are available as appendices.